



Universidade Federal do ABC

Simulation Exercise #06: Simulation of Temporal Systems

INF301 – System Modeling and Simulation – W06

Prof. Luiz Henrique Bonani do Nascimento

Universidade Federal do ABC

Report Guidelines

Use a notebook on the Google Collaboratory platform to generate a report containing any explanations and comments you deem relevant, along with your codes and figures.

The graphs in the figures should be **self-explanatory**, with axis names and data captions. Use a **font size appropriate** for presentation in a document.

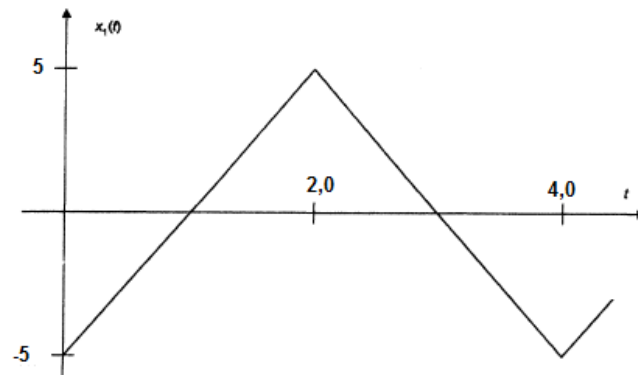
The language to be used is Python. However, **the use of pre-built Python libraries is not permitted**, except for those used in the examples.

Submit a **single notebook file in ipynb format**, with the file name in the format SEON_NameSurname.ipynb, where N is the SE number, Name is your first name, and Surname is your last name.

Remember that plagiarism will not be tolerated under any circumstances!

Problem 1 (20 points)

Use relational notation and construct a function to describe the periodic signal shown below. Analyze the results and provide the proper comments.



Problem 2 (40 points)

Suppose the signal described in Problem 01 is shifted by the addition of a 5 V DC level. Furthermore, a uniformly distributed noise is added in the interval $[-3, 3]$. The resulting signal is then subjected to a low-pass filter with time constant τ .

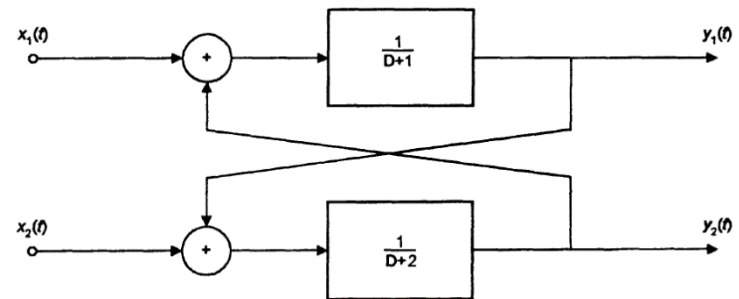
Find a good value for τ by testing time constant values of 10; 1; 0.1; and 0.01 (you can also try other values). Use a step of 0.01 for the simulation. Analyze the results and provide the proper comments.

Problem 3 (40 points)

Simulate the system, in the time interval between 0 and 15 seconds.

Data: deterministic input signals

$x_1(t) = \sin(0,5 t)$ and $x_2(t) = \cos(0,5 t)$. The initial conditions are zero.



- Show the results of the system simulation for $y_1(t)$ and $y_2(t)$.
- Repeat the system simulation considering the addition of uniform noise with zero mean and variance $1/3$ to both inputs $x_1(t)$ and $x_2(t)$.
- Pass the noisy outputs through low-pass filters with time constants $\tau = 0.5; 0.8$, and 2 .
- Provide comments on the achieved results.